1

Introduction to Computer Graphics

1.1 Introduction

The computer is an information processing machine. It is a tool for storing, manipulating and correlating data. There are many ways to communicate the processed information to the user. The computer graphics is one of the most effective and commonly used way to communicate the processed information to the user. It displays the information in the form of graphics objects such as pictures, charts, graphs and diagrams instead of simple text. Thus we can say that computer graphics makes it possible to express data in pictorial form. The picture or graphics object may be an engineering drawing, business graphs, architectural structures, a single frame from an animated movie or a machine parts illustrated for a service manual. It is the fundamental cohesive concept in computer graphics. Therefore, it is important to understand -

- How pictures or graphics objects are presented in computer graphics ?
- How pictures or graphics objects are prepared for presentation ?
- How previously prepared pictures or graphics objects are presented ?
- How interaction with the picture or graphics object is accomplished ?

In computer graphics, pictures or graphics objects are presented as a collection of discrete picture elements called pixels. The pixel is the smallest addressable screen element. It is the smallest piece of the display screen which we can control. The control is achieved by setting the intensity and colour of the pixel which compose the screen. This is illustrated in Fig. 1.1.

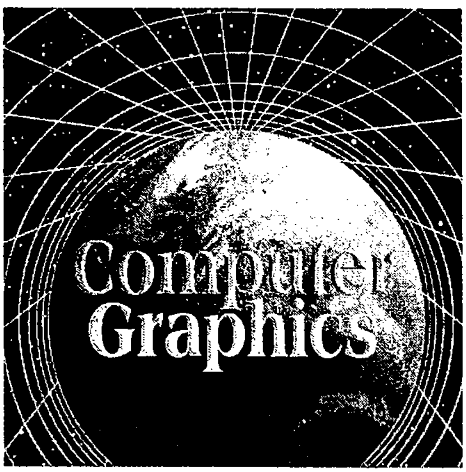


Fig. 1.1 Representation of picture

Each pixel on the graphics display does not represent mathematical point. Rather, it represents a region which theoretically can contain an infinite number of points. For example, if we want to display point P_1 whose coordinates are (4.2, 3.8) and point P_2

whose coordinates are (4.8, 3.1) then P_1 and P_2 are represented by only one pixel (4, 3), as shown in the Fig. 1.2. In general, a point is represented by the integer part of x and integer part of y, i.e., pixel (int (x), int (y)).

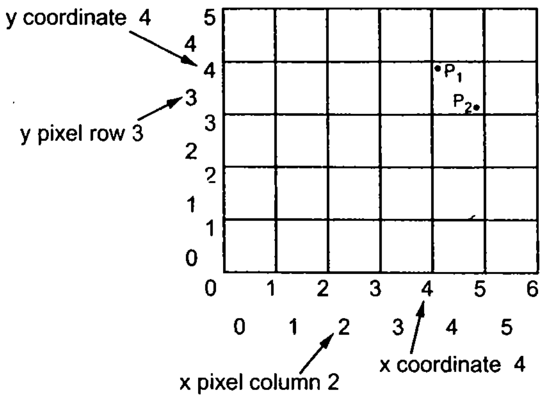


Fig. 1.2 Pixel display area of 6 × 5

The special procedures determine which pixel will provide the best approximation to the desired picture or graphics object. The process of determining the appropriate pixels for representing picture or graphics object is known as **rasterization**, and the process of representing continuous picture or graphics object as a collection of discrete pixels is called **scan conversion**.

The computer graphics allows rotation, translation, scaling and performing various projections on the picture before displaying it. It also allows to add effects such as hidden surface removal, shading or transparency to the picture before final representation. It provides user the control to modify contents, structure, and appearance of pictures or graphics objects using input devices such as a keyboard, mouse, or touch-sensitive panel on the screen. There is a close relationship between the input devices and display devices. Therefore, graphics devices includes both input devices and display devices.

1.2 Image Processing as Picture Analysis

The computer graphics is a collection, combination and representation of real or imaginary objects from their computer-based models. Thus we can say that computer graphics concerns the pictorial synthesis of real or imaginary objects. However, the related field image processing or sometimes called picture analysis concerns the analysis of scenes, or the reconstruction of models of 2D or 3D objects from their picture. This is exactly the reverse process. The image processing can be classified as

- Image enhancement
- Pattern detection and recognition
- Scene analysis and computer vision

The image enhancement deals with the improvement in the image quality by eliminating noise or by increasing image contrast. Pattern detection and recognition deal with the detection and clarification of standard patterns and finding deviations from these patterns. The optical character recognition (OCR) technology is an practical example of pattern detection and recognition. Scene analysis and computer vision deals with the recognition and reconstruction of 3D model of scene from several 2D images.

The above three fields of image processing proved their importance in many area such as finger print detection and recognition, modeling of buildings, ships, automobiles etc., and so on.

We have discussed the two fields : computer graphics and image processing of computer processing of pictures. In the initial stages they were quite separate disciplines. But now a days they use some common features, and overlap between them is growing. They both use raster displays.

1.3 The Advantages of Interactive Graphics

Let us discuss the advantages of interactive graphics.

- Today, a high quality graphics displays of personal computer provide one of the most natural means of communicating with a computer.
- It provides tools for producing pictures not only of concrete, "real-world" objects but also of abstract, synthetic objects, such as mathematical surfaces in 4D and of data that have no inherent geometry, such as survey results.
- It has an ability to show moving pictures, and thus it is possible to produce animations with interactive graphics.
- With interactive graphics use can also control the animation by adjusting the speed, the portion of the total scene in view, the geometric relationship of the objects in the scene to one another, the amount of detail shown and so on.
- The interactive graphics provides tool called **motion dynamics**. With this tool user can move and tumble objects with respect to a stationary observer, or he can make objects stationary and the viewer moving around them. A typical example is walk throughs made by builder to show flat interior and building surroundings. In many case it is also possible to move both objects and viewer.
- The interactive graphics also provides facility called **update dynamics**. With update dynamics it is possible to change the shape, colour or other properties of the objects being viewed.
- With the recent development of digital signal processing (DSP) and audio synthesis chip the interactive graphics can now provide audio feedback alongwith the graphical feedbacks to make the simulated environment even more realistic.

In short, interactive graphics permits extensive, high-bandwidth user-computer interaction. It significantly enhances the ability to understand information, to perceive trends and to visualize real or imaginary objects either moving or stationary in a realistic environment. It also makes it possible to get high quality and more precise results and products with lower analysis and design cost.

1.4 Applications of Computer Graphics

The use of computer graphics is wide spread. It is used in various areas such as industry, business, government organisations, education, entertainment and most recently the home. Let us discuss the representative uses of computer graphics in brief.

- **User Interfaces :** User friendliness is one of the main factors underlying the success and popularity of any system. It is now a well established fact that graphical interfaces provide an attractive and easy interaction between users and computers. The built-in graphics provided with user interfaces use visual control items such as buttons, menus, icons, scroll bar etc, which allows user to interact with computer only by mouse-click. Typing is necessary only to input text to be stored and manipulated.
- **Plotting of graphics and chart :** In industry, business, government and educational organisations, computer graphics is most commonly used to create 2D and 3D graphs of mathematical, physical and economic functions in form of histograms, bars and pie-charts. These graphs and charts are very useful for decision making.
- **Office automation and Desktop Publishing :** The desktop publishing on personal computers allow the use of graphics for the creation and dissemination of information. Many organisations does the in-house creation and printing of documents. The desktop publishing allows user to create documents which contain text, tables, graphs and other forms of drawn or scanned images or pictures. This is one approach towards the office automation.
- **Computer-aided Drafting and Design :** The computer-aided drafting uses graphics to design components and systems electrical, mechanical, electromechanical and electronic devices such as automobile bodies, structures of building, airplane, ships, very large-scale integrated (VLSI) chips, optical systems and computer networks.
- **Simulation and Animation :** Use of graphics in simulation makes mathematic models and mechanical systems more realistic and easy to study. The interactive graphics supported by animation software proved their use in production of animated movies and cartoons films.
- **Art and Commerce :** There is a lot of development in the tools provided by computer graphics. This allows user to create artistic pictures which express messages and attract attentions. Such pictures are very useful in advertising.
- **Process Control :** By the use of computer now it is possible to control various processes in the industry from a remote control room. In such cases, process systems and processing parameters are shown on the computer with graphic symbols and identifications. This makes it easy for operator to monitor and control various processing parameters at a time.
- **Cartography :** Computer graphics is also used to represent geographic maps, weather maps, oceanographic charts, contour maps, population density maps and so on.

1.5 Classification of Applications

In the last section we have seen various uses of computer graphics. These uses can be classified as shown in the Fig. 1.3. As shown in the Fig. 1.3, the use of computer graphics can be classified according to dimensionality of the object to be drawn : 2D or 3D. It can also be classified according to kind of picture : Symbolic or Realistic. Many computer graphics applications are classified by the type of interaction. The type of interaction determines the user's degree of control over the object and its image. In controllable interaction user can change the attributes of the images. Role of picture gives the another classification. Computer graphics is either used for representation or it can be an end product such as drawings. Pictorial representation gives the final classification of use computer graphics. It classifies the use of computer graphics to represent pictures such as line drawing, black and white, colour and so on.

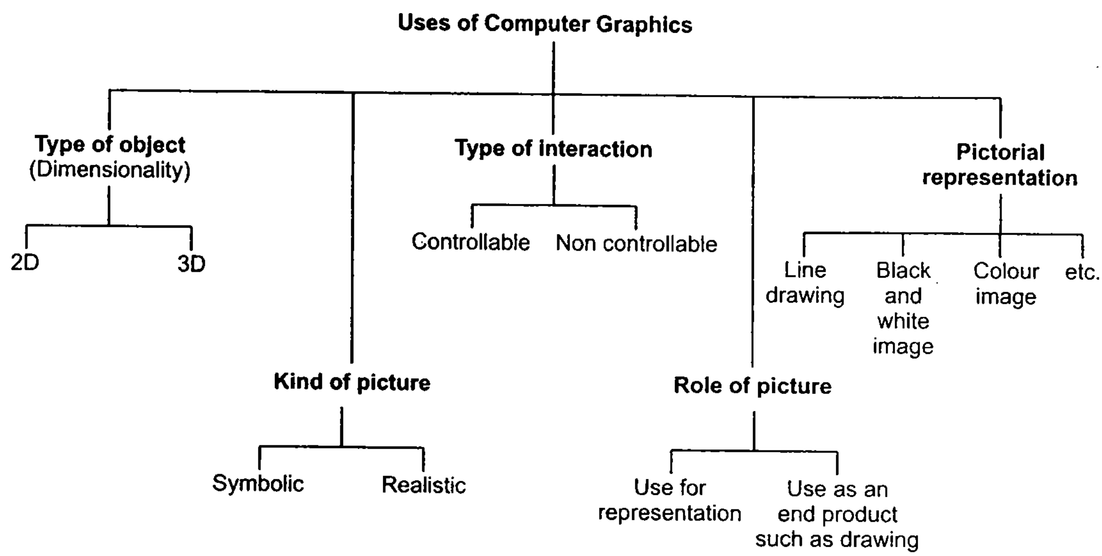


Fig. 1.3

1.6 Input Devices

Number of devices are available for data input in the graphics systems. These include keyboard, mouse, trackball, spaceball, joystick, digitizers, scanners and so on. Let us discuss them.

1.6.1 Keyboard

The keyboard is a primary input device for any graphics system. It is used for entering text and numbers, i.e. on graphics data associated with pictures such as labels x-y coordinates etc.

Keyboards are available in various sizes, shapes and styles. Fig. 1.4 shows standard keyboard. It consists of

- Alphanumeric key
- Function keys
- Modifier keys
- Cursor movement keys
- Numeric keypad

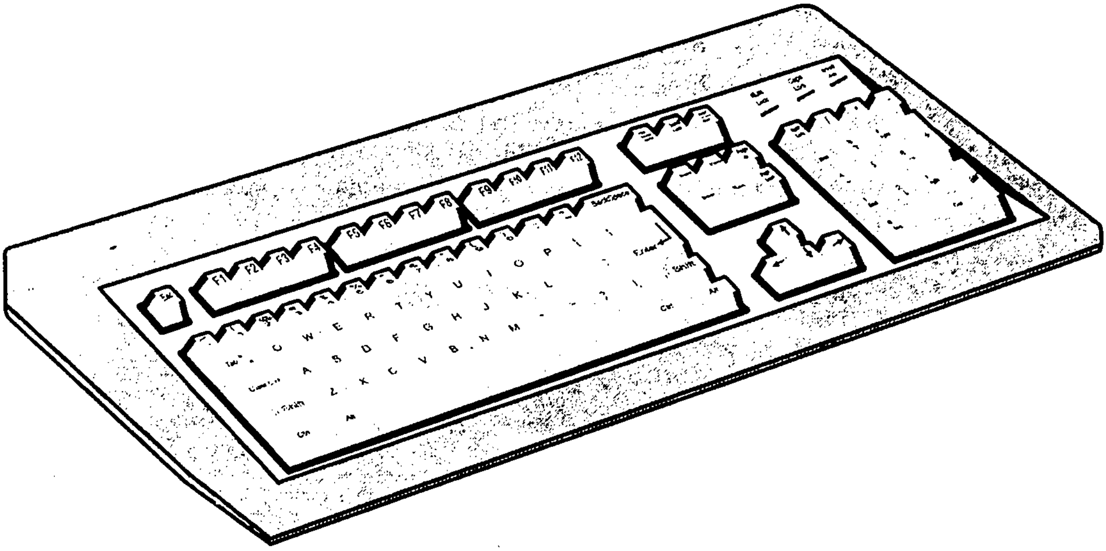


Fig. 1.4

When we press a key on the keyboard, keyboard controller places a code corresponding to key pressed into a part of its memory, called **keyboard buffer**. This code is called **scan code**. The keyboard controller informs CPU of the computer about the key press with the help of an interrupt signal. The CPU then reads the scan code from the keyboard buffer, as shown in the Fig. 1.5.

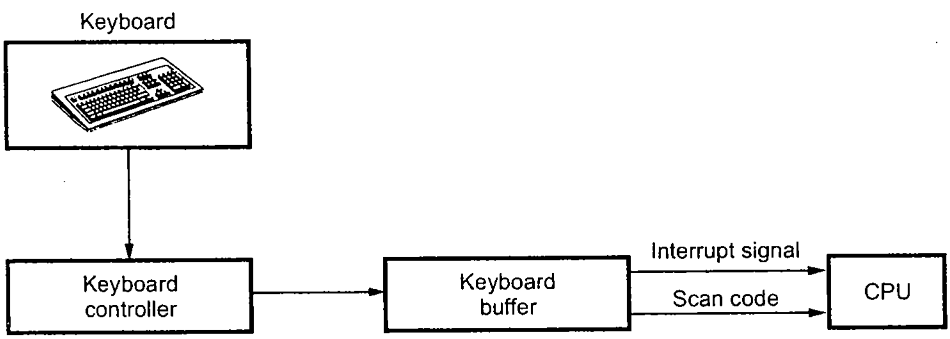


Fig. 1.5 Getting the scan code from keyboard

1.6.2 Mouse

A mouse is a palm-sized box used to position the screen cursor. It consists of ball on the bottom connected to wheels or rollers to provide the amount and direction of movement. One, two or three buttons are usually included on the top of the mouse for signaling the execution of some operation. Now-a-days mouse consists of one more wheel on the top to scroll the screen pages.

(a) Mouse (b) Mouse with scrolling wheel

Fig. 1.6

1.6.3 Trackball and Spaceball

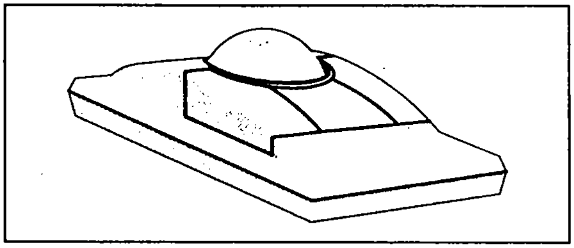


Fig. 1.7 Trackball

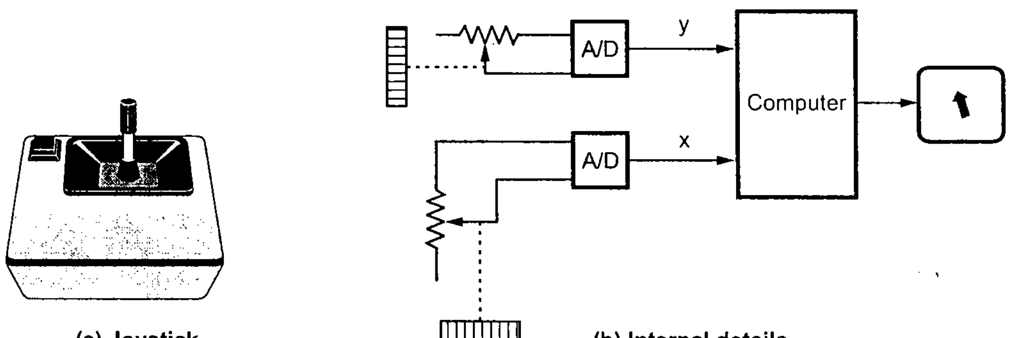
As the name implies, a trackball is a ball that can be rotated with the fingers or palm of the hand to produce screen cursor movement. The potentiometers attached to the trackball are used to measure the amount and direction of rotation.

The trackball is a two dimensional positioning device whereas spaceball provides six degree of freedom. It does not actually move. It consists of strain guages which measure the amount of pressure applied to the spaceball to provide input for spatial positioning and orientation as the ball is pushed or pulled in various directions. It is usually used in three-dimensional positioning and selecting operations in virtual-reality systems.

1.6.4 Joysticks

A joystick has a small, vertical lever (called the stick) mounted on the base and used to steer the screen cursor around. It consists of two potentiometers attached to a single lever. Moving the lever changes the settings on the potentiometers. The left or right movement is indicated by one potentiometer and forward or back movement is indicated by other potentiometer. Thus with a joystick both x and y-coordinate positions can be simultaneously altered by the motion of a single lever. This is illustrated in Fig. 1.8.

Some joysticks may return to their zero (center) position when released. Joysticks are inexpensive and are quite commonly used where only rough positioning is needed.



(a) Joystick (b) Internal details

Fig. 1.8

1.6.5 Data Glove

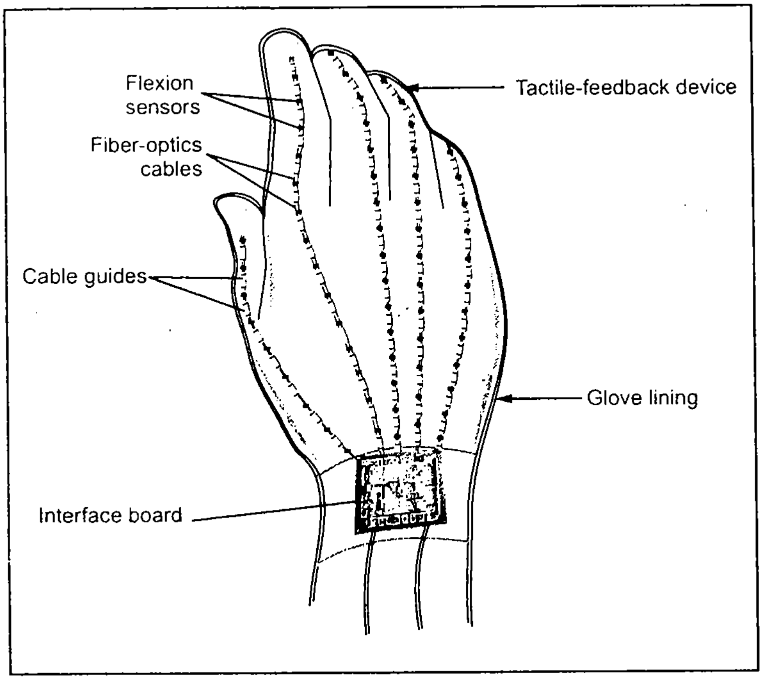


Fig. 1.9

The data glove is used to grasp a virtual object. The Fig. 1.9 shows the data glove. It is constructed with a series of sensors that detect hand and finger motions. Each sensor is a short length of fiberoptic cable, with a light-emitting diode (LED) at one end and a phototransistor at the other end. The surface of a cable is roughened in the area where it is to be sensitive to bending. When the cable is flexed, some of the LED's light is lost, so less light is received by the phototransistor.

The input from the glove can be used to position or manipulate objects in a virtual scene. Thus by wearing the dataglove, a user can grasp, move and rotate objects and then release them.

1.6.6 Digitizer/Graphical Tablet

For applications such as tracing we need a device called a **digitizer** or a **graphical tablet**. It consists of a flat surface, ranging in size from about 6 by 6 inches up to 48 by 72 inches or more, which can detect the position of a movable stylus. Fig. 1.10 shows a small tablet with penlike stylus.

Different graphics tablets use different techniques for measuring position, but they all resolve the position into a horizontal and a vertical direction, which correspond to the axes of the display. Most graphics tablets use an electrical sensing mechanism to determine the position of the stylus. In one such arrangement, a grid of wire on 1/4 to 1/2 inch centers is embedded in the tablet surface. Electromagnetic signals generated by electrical pulses applied in sequence to the wires in the grid induce an electrical signal in a wire coil in the stylus. The strength of the signal induced by each pulse

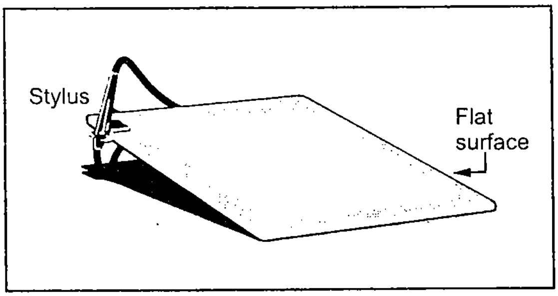


Fig. 1.10

is used to determine the position of the stylus. The signal strength is also used to determine roughly how far the stylus or cursor is from the graphical tablet.

Every time user may not wish to enter stylus position into the computer. In such cases user can lift the stylus or make the tablet off by pressing a switch provided on the stylus.

Other graphical tablet technologies use sound (sonic) coupling and resistive coupling. The sonic tablet uses sound waves to couple the stylus to microphones positioned on the periphery of the digitizing area. Sound brust are created by an electrical spark at the tip of the stylus. The time between when the spark occurs and when its sound arrives at each microphone is proportional to the distance from the stylus to each microphone. The sonic tablets mainly used in 3D positioning the devices. The resistive tablet uses a battery powered stylus that emits high-frequency radio signals. The tablet is a piece of glass coated with a thin layer of conducting material in which an electrical potential is induced by the radio signals. The strength of the signals at the edges of the tablet is inversely proportional to the distance to the stylus and can thus be used to calculate the stylus position.

1.6.7 Image Scanners

The scanner is a device, which can be used to store drawing, graphs, photos or text available in printed form for computer processing. The scanners use the optical scanning mechanism to scan the information. The scanner records the gradation of gray scales or colour and stores them in the array. Finally, it stores the image information in a specific file format such as JPEG, GIF, TIFF, BMP and so on. Once the image is scanned, it can be processed or we can apply transformations to rotate, scale, or crop the image using image processing softwares such as photo-shop or photo-paint. Scanners are available in variety of sizes and capabilities.

Fig. 1.11 shows the working of photoscanner. As shown in the Fig. 1.11, the photograph is mounted on a rotating drum. A finely collimated light beam is directed at the photo, and the amount of light reflected is measured by a photocell. As the drum rotates, the light source slowly moves from one end to the other, thus doing a raster scan of the entire photograph.

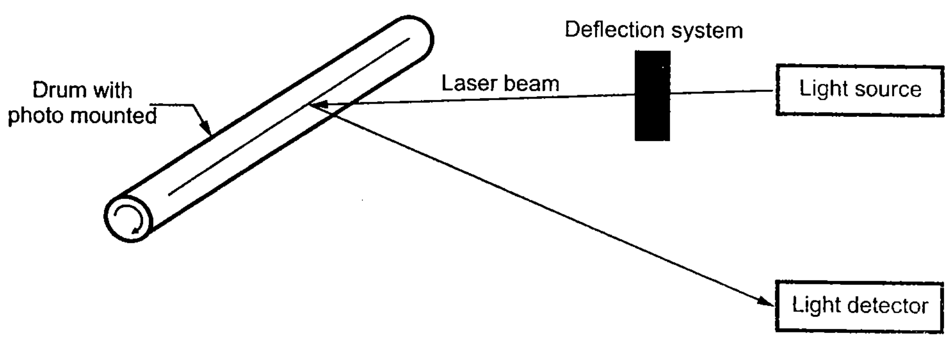


Fig. 1.11 Photoscanner

For coloured photographs, multiple passes are made, using filters in the front of the photocell to separate out various colours.

Other type of scanners are electro-optical devices that use arrays of light sensitive charge coupled devices (CCDs) to turn light reflected from, or transmitted through, artwork, photographs, slides etc. into a usable digital file composed of pixel information.

The optical resolution and colour depth are the two important specifications of the scanner. The photoscanners have resolution upto 2000 units per inch. Resolution of the CCD array is 200 to 1000 units per inch which is less than the photoscanners. The colour depth is expressed in bits. It specifies the number of colours scanner can capture.

According to construction the scanners also can be classified as :

Flatbed scanners, also called desktop scanners, are the most versatile and commonly used scanners. In fact, this article focuses on the technology as it relates to flatbed scanners.

Sheet-fed scanners are similar to flatbed scanners except the document is moved and the scan head is immobile. A sheet-fed scanner looks a lot like a small portable printer.

Handheld scanners use the same basic technology as a flatbed scanner, but relay on the user to move them instead of a motorized belt. This type of scanner typically does not provide good image quality. However, it can be useful for capturing an image quickly.

Drum scanners are used by the publishing industry to capture incredibly detailed images. They use a technology called a **photomultiplier tube** (PMT). In PMT, the document to be scanned is mounted on a glass cylinder. Located at the center of the cylinder is a sensor that splits light bounced from the document into three beams. Each beam is sent through a colour filter into a photomultiplier tube where the light is changed into an electrical signal.

1.6.8 Touch Panels

As the name implies, touch panels allow displayed objects or screen positions to be selected with the touch of a finger. The touch panels are the transparent devices which are fitted on the screen. They consist of touch sensing mechanism. Touch input can be recorded using optical, electrical or acoustical methods.

Optical touch panels use a line of infrared light emitting diodes along one vertical edge and along one horizontal edge of the screen. The opposite vertical and horizontal edges contain light detectors. These detectors are used to record which beams are interrupted when the panel is touched. The two crossing beams that are interrupted indicate the horizontal and vertical coordinates of the screen position selected.

In electrical touch panels, two transparent plates are used. These plates are separated by a small distance. One plate is coated with conducting material and other plate is coated with resistive material. When outer plate is touched, it is forced to contact with the inner plate. This contact creates a voltage drop across the resistive plate that is used to determine the coordinate values of the selected screen position.

An acoustic touch panels use high frequency sound waves in horizontal and vertical directions across a glass plate. Touching the screen causes partial reflection of each wave from finger to the emitter. The screen coordinates of point of contact are then calculated by measuring time between the transmission of each wave and its reflection to the emitter.

1.6.9 Light Pens

Light pen is a pencil shaped device used to select positions by detecting the light coming from points on the CRT screen. It consists of a photoelectric cell housed in a pencil like case as shown in the Fig. 1.12.

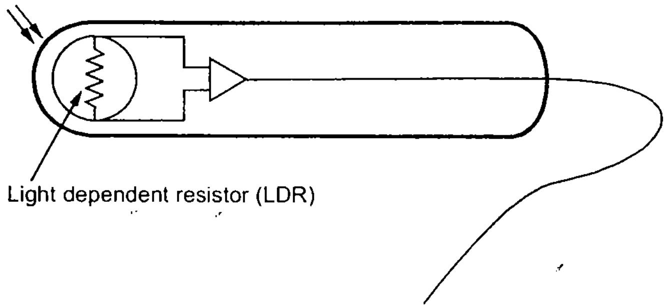


Fig. 1.12

The light pen may be pointed at the screen. An optical system focuses light on to the photo cell in the field view of the pen. The pen will send a pulse whenever the phosphor below it is illuminated. This output of the photo-cell is then amplified and carried over a shielded cable to light pen interface. A 'detect' by the light pen can either be used to cause an immediate interrupt to the computer through an interface or be used to set a flip-flop which is cleared when read by the computer. Thus, the system can identify the part of the graphics to which the pen is pointing. It records the co-ordinate position of the electron beam.

Light pen is an event driven device. After displaying each point, one can test the light pen flip-flop. Thus exact location of the spot to which the light pen is pointing can be detected. For using light pen for positioning, a tracking program is run on the computer. The display processor also has the capability to disable the light pen during the refresh cycle. This ensures the inputs not desired by the operator are ignored. To facilitate the operation of the light pen, a finger operated switch is provided to control the light reaching the photo cell.

The response time of the pen is also important. For slow displays transistor type photo cells such as photo diodes are used. These are small, inexpensive and suitable for hand held operation. The response time is about one microsecond. For the highspeed displays photo-multiplier tube is used. It is bulky and uses fiber-optic cable.

Now a days improved light pens are available. These consist of a matrix of fibre optic sensors.

1.6.10 Voice Systems

Speech recognizers are used in some graphics systems as input devices to accept voice commands. Such a voice-system input can be used to initiate graphics operations or to enter data. These systems operate by matching an input with a predefined dictionary of words and phrases.

The voice recognizers are classified accordings to whether or not they must be trained to recognize the waveforms of a particular speaker, and whether they can recognize connected speech as opposed to single words or phrases. The speaker independent recognizers have very limited vocabularies. Usually, they include only the digits and 50 to 100 words.

The Fig. 1.13 shows the typical voice recognition system. In such systems microphone is also included to minimize input of other backgrond sounds.

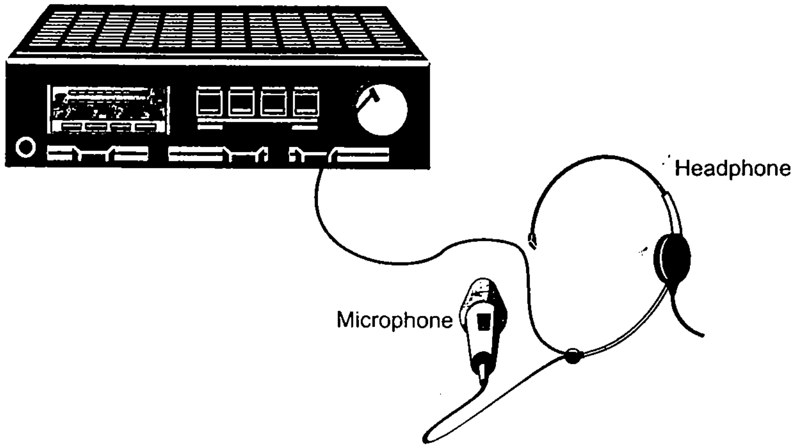


Fig. 1.13

The one advantage of voice system over other devices is that in voice systems the attention of the operator does not have to be switched from one device to another to enter a command.

1.7 Output Devices

The output devices can be classified as display devices and hardcopy devices. Let us see some of them.

1.7.1 Video Display Devices

The most commonly used output device in a graphics system is a video monitor. The operation of most video monitors is based on the standard **cathode-ray-tube** (CRT) design. Let us see the basics of the CRT.

1.7.1.1 Cathode-Ray-Tubes

A CRT is an evacuated glass tube. An electron gun at the rear of the tube produces a beam of electrons which is directed towards the front of the tube (screen). The inner side of the screen is coated with phosphor substance which gives off light when it is stroked by electrons. This is illustrated in Fig. 1.14. It is possible to control the point at which the electron beam strikes the screen, and therefore the position of the dot upon the screen, by deflecting the electron beam. The Fig. 1.15 shows the electrostatic deflection of the electron beam in a CRT.

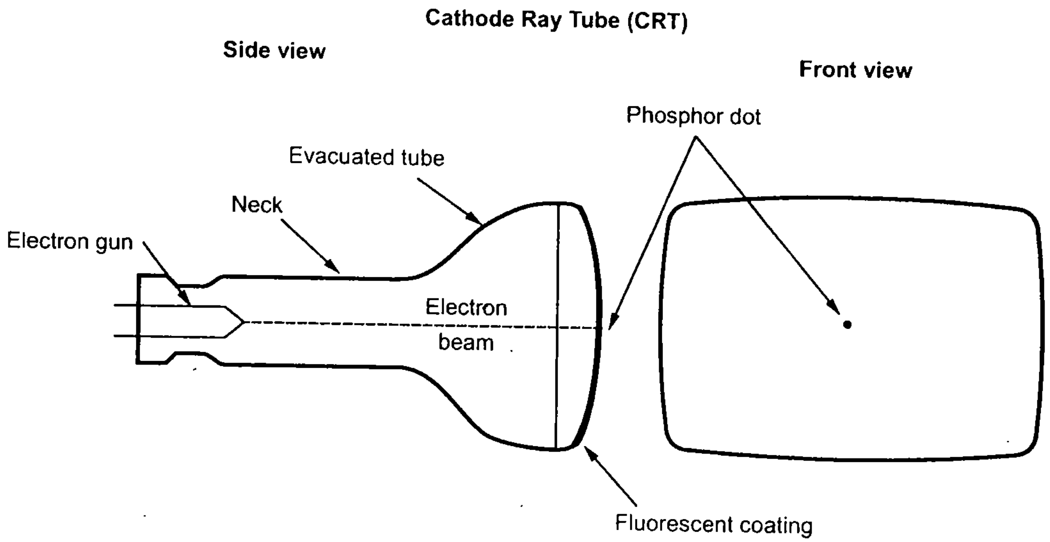


Fig. 1.14 Simplified representation of CRT

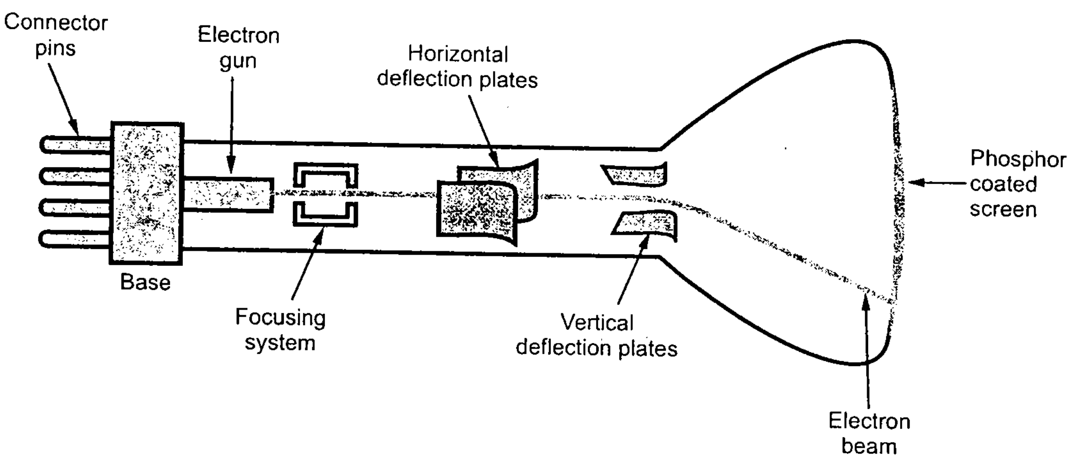


Fig. 1.15 Cathode Ray Tube

The deflection system of the cathode-ray-tube consists of two pairs of parallel plates, referred to as the vertical and horizontal deflection plates. The voltage applied to vertical plates controls the vertical deflection of the electron beam and voltage applied to the horizontal deflection plates controls the horizontal deflection of the electron beam. There are two techniques used for producing images on the CRT screen : Vector scan/random scan and Raster scan.

1.7.1.2 Vector Scan/Random Scan Display

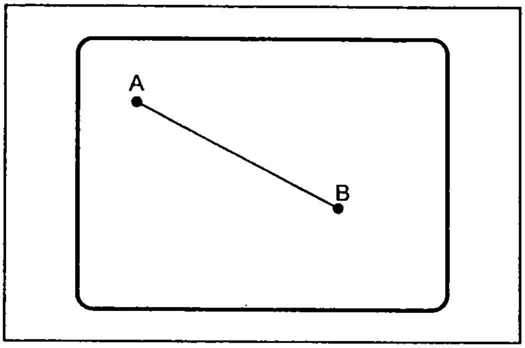


Fig. 1.16 Vector scan CRT

As shown in Fig. 1.16, vector scan CRT display directly traces out only the desired lines on CRT i.e. If we want a line connecting point A with point B on the vector graphics display, we simply drive the beam deflection circuitry, which will cause beam to go directly from point A to B. If we want to move the beam from point A to point B without showing a line between points, we can blank the beam as we move it. To move the beam across the CRT, the information about both, magnitude and direction is required. This information is generated with the help of vector graphics generator.

The Fig. 1.17 shows the typical vector display architecture. It consists of display controller, Central Processing Unit (CPU), display buffer memory and a CRT. A display controller is connected as an I/O peripheral to the central processing unit (CPU). The display buffer memory stores the computer produced display list or display program. The program contains point and line plotting commands with (x, y) or (x, y, z) end point coordinates, as well as character plotting commands. The display controller interprets commands for plotting points, lines and characters and sends digital and point coordinates to a vector generator. The vector generator then converts the digital coordinate values to analog voltages for beam-deflection circuits that displace an electron beam writing on the CRT's phosphor coating.

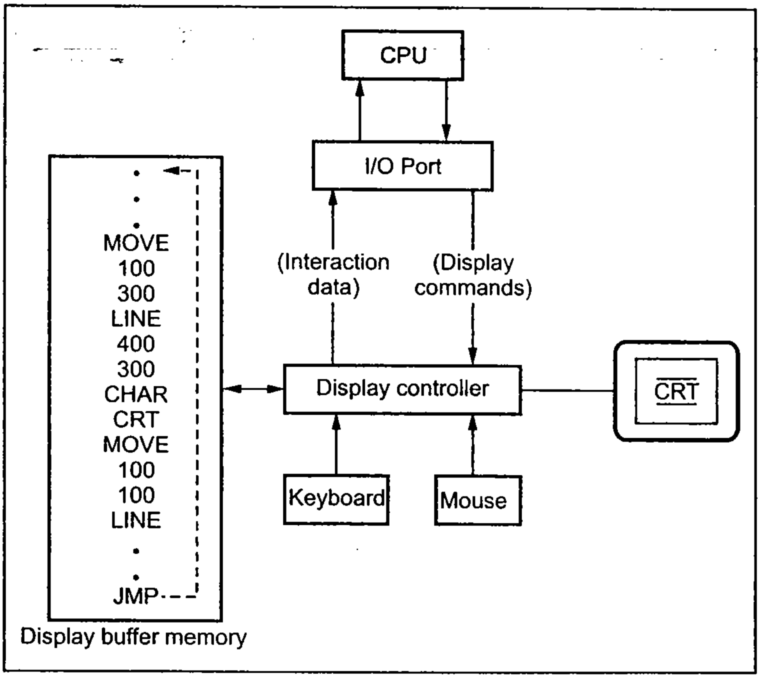


Fig. 1.17 Architecture of a vector display

In vector displays beam is deflected from end point to end point, hence this technique is also called **random scan**. We know as beam, strikes phosphor it emits light. But phosphor light decays after few milliseconds and therefore it is necessary to repeat through the display list to refresh the phosphor at least 30 times per second to avoid flicker. As display buffer is used to store display list and it is used for refreshing, the display buffer memory is also called **refresh buffer**.

1.7.1.3 Raster Scan Display

The Fig. 1.18 shows the architecture of a raster display. It consists of display controller, central processing unit (CPU), video controller, refresh buffer, keyboard, mouse and the CRT.

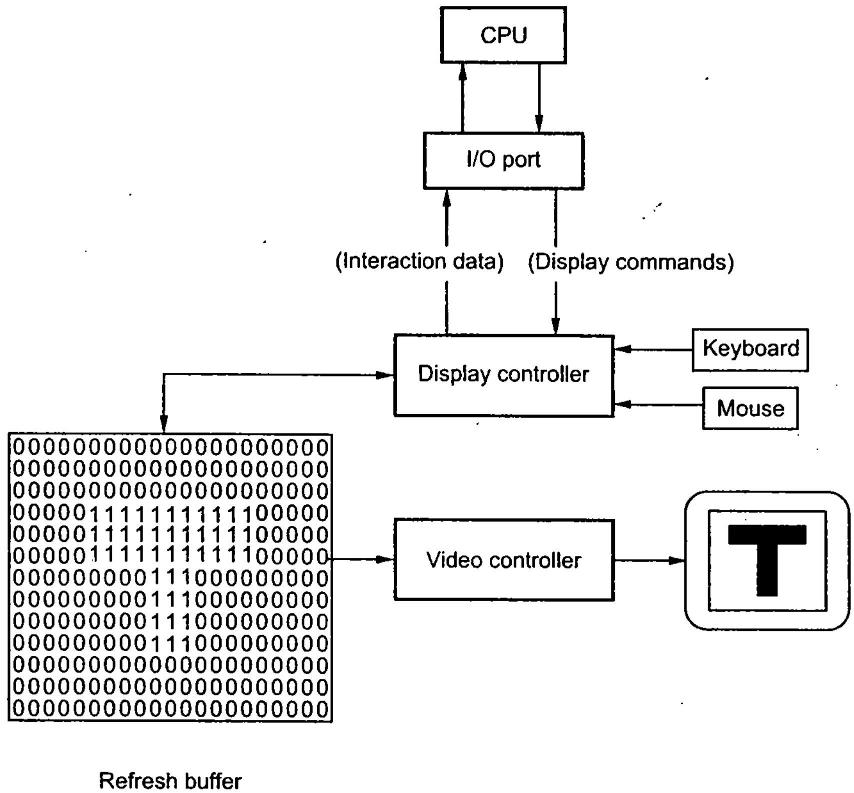


Fig. 1.18 Architecture of a raster display

As shown in the Fig. 1.18, the display image is stored in the form of 1s and 0s in the refresh buffer. The video controller reads this refresh buffer and produces the actual image on the screen. It does this by scanning one scan line at a time, from top to bottom and then back to the top, as shown in the Fig. 1.18.

Raster scan is the most common method of displaying images on the CRT screen. In this method, the horizontal and vertical deflection signals are generated to move the beam all over the screen in a pattern shown in the Fig. 1.19

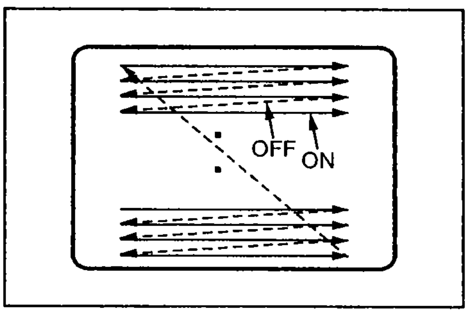


Fig. 1.19 Raster scan CRT

Here, the beam is swept back and forth from the left to the right across the screen. When the beam is moved from the left to the right, it is ON. The beam is OFF, when it is moved from the right to the left as shown by dotted line in Fig. 1.19

When the beam reaches the bottom of the screen, it is made OFF and rapidly retraced back to the top left to start again. A display produced in this way is called **raster scan display**. Raster scanning process is similar to reading different lines on the page of a book. After completion of scanning of one line, the electron beam flies back to the start of next line and process repeats. In the raster scan display, the screen image is maintained by repeatedly scanning the same image. This process is known as **refreshing of** screen.

In raster scan displays a special area of memory is dedicated to graphics only. This memory area is called **frame buffer.** It holds the set of intensity values for all the screen points. The stored intensity values are retrieved from frame buffer and displayed on the screen one row (scan line) at a time. Each screen point is referred to as a **pixel** or **pel** (shortened forms of picture element). Each pixel on the screen can be specified by its row and column number. Thus by specifying row and column number we can specify the **pixel position** on the screen.

Intensity range for pixel positions depends on the capability of the raster system. It can be a simple black and white system or colour system. In a simple black and white system, each pixel position is either on or off, so only one bit per pixel is needed to control the intensity of the pixel positions. Additional bits are required when colour and intensity variations can be displayed. Upto 24 bits per pixel are included in high quality display systems, which can require several megabytes of storage space for the frame buffer. On a black and white system with one bit per pixel, the frame buffer is commonly called a **bitmap**. For systems with multiple bits per pixel, the frame buffer is often referred to as a **pixmap**.

Vector Scan Display	Raster Scan Display
1. In vector scan display the beam is moved between the end points of the graphics primitives.	1. In raster scan display the beam is moved all over the screen one scan line at a time, from top to bottom and then back to top.
2. Vector display flickers when the number of primitives in the buffer becomes too large.	2. In raster display, the refresh process is independent of the complexity of the image.
3. Scan conversion is not required.	3. Graphics primitives are specified in terms of their endpoints and must be scan converted into their corresponding pixels in the frame buffer.
4. Scan conversion hardware is not required.	4. Because each primitive must be scan-converted, real time dynamics is far more computational and requires separate scan conversion hardware.
5. Vector display draws a continuous and smooth lines.	5. Raster display can display mathematically smooth lines, polygons, and boundaries of curved primitives only by approximating them with pixels on the raster grid.
6. Cost is more.	6. Cost is low.
7. Vector display only draws lines and characters.	7. Raster display has ability to display areas filled with solid colours or patterns.

Table 1.1

Display File Interpreter

We have seen that display file contains information necessary to construct the picture. This information is in the form of commands. The program which converts these commands into actual picture is called display file interpreter. It is an interface between graphics representation in the display file and the display device, as shown in the Fig. 1.25.

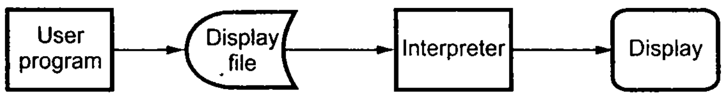


Fig. 1.25 Display file and interpreter

As shown the Fig. 1.25, the display process is divided into two steps : first the image is stored in the display file structure and then it is interpreted by an appropriate interpreter to get the actual image.

Instead of this, if we store actual image for particular display device it may not run on other displays. To achieve the device independence the image is stored in the raw format, i.e. in the display file format and then it is interpreted by an appropriate interpreter to run on required display.

Another advantage of using interpreter is that saving raw image takes much less storage than saving the picture itself.

Display Controller

In some graphics systems a separate computer is used to interpret the commands in the display file. Such computer is known as display controller. Display controller access display file and it cycles through each command in the display file once during every refresh cycle Fig. 1.26 shows the vector scan system with display controller.

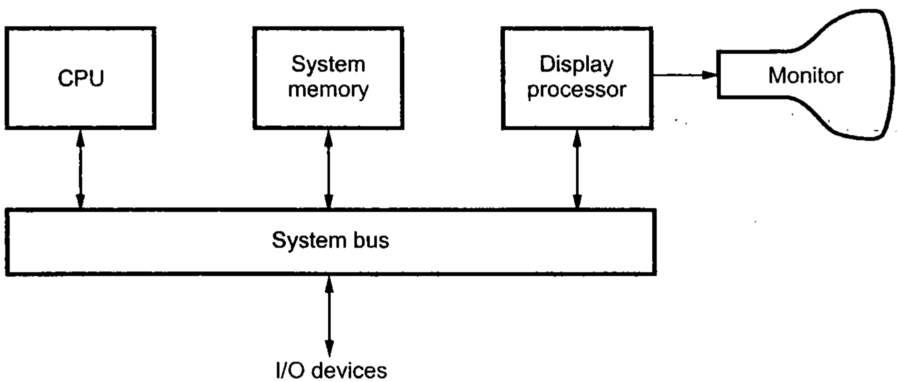


Fig. 1.26 Vector scan system

In the raster scan display systems, the purpose of display controller is to free the CPU from the graphics routine task. Here, display controller is provided with separate memory area as shown in the Fig. 1.26. The main task of display controller is to digitize a picture definition given in an application program into a set of pixel-intensity values for storage in the frame buffer. This digitization process is known as **scan conversion**.

Display controller are also designed to perform a number of additional operations. These operations include

- Generating various line styles (dashed, dotted, or solid)
- Display colour areas
- Performing certain transformations and
- Manipulations on displayed objects

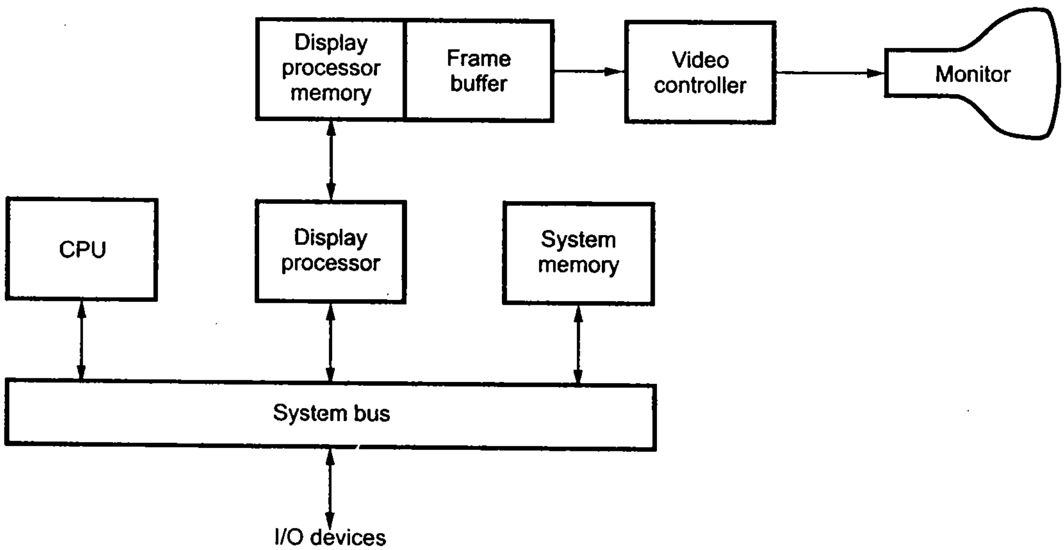


Fig. 1.27 Raster scan system with a display processor

1.7.1.4 Colour CRT Monitors

A CRT monitor displays colour pictures by using a combination of phosphors that emit different-coloured light. It generates a range of colours by combining the emitted light from the different phosphors. There are two basic techniques used for producing colour displays :

- Beam-penetration technique and
- Shadow-mask technique

Beam-penetration Technique

This technique is used with random-scan monitors. In this technique, the inside of CRT screen is coated with two layers of phosphor, usually red and green. The displayed colour depends on how far the electron beam penetrates into the phosphor layers. The outer layer is of red phosphor and inner layer is of green phosphor. A beam of slow electrons excites only the outer red layer. A beam of very fast electrons penetrates through the red layer and excites the inner green layer. At intermediate beam speeds, combinations of red and green light are emitted and two additional colours, orange and yellow displayed. The beam acceleration voltage controls the speed of the electrons and hence the screen colour at any point on the screen.

Merits and Demerits

- It is an inexpensive technique to produce colour in random scan monitors.
- It can display only four colours.
- The quality of picture produced by this technique is not as good as compared to other techniques.

Shadow Mask Technique

The shadow mask technique produces a much wider range of colours than the beam penetration technique. Hence this technique is commonly used in raster-scan displays including colour TV. In a shadow mask technique, CRT has three phosphor colour dots at each pixel position. One phosphor dot emits a red light, another emits a green light, and the third emits a blue light. The Fig. 1.28 shows the shadow mask CRT. It has three electron guns, one for each colour dot, and a shadow mask grid just behind the phosphor coated screen.

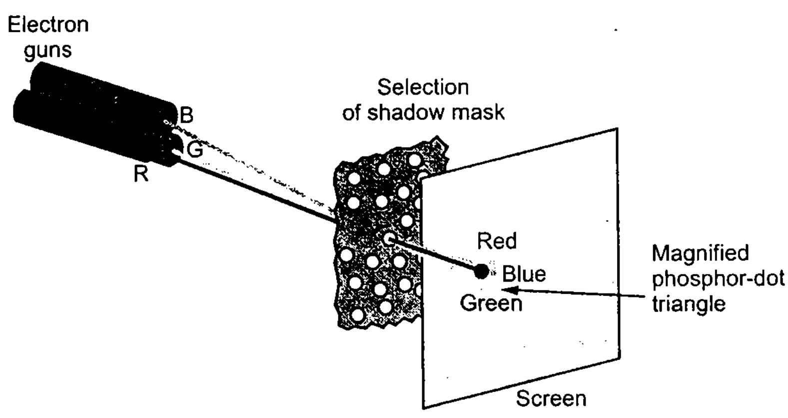


Fig. 1.28

The shadow mask grid consists of series of holes aligned with the phosphor dot pattern. As shown in the Fig. 1.28, three electron beams are deflected and focused as a group onto the shadow mask and when they pass through a hole in the shadow mask, they excite a dot triangle. A dot triangle consists of three small phosphor dots of red, green and blue colour. These phosphor dots are arranged so that each electron beam can activate only its corresponding colour dot when it passes through the shadow mask. A dot triangle when activated appears as a small dot on the screen which has colour of combination of three small dots in the dot triangle. By varying the intensity of the three electron beams we can obtain different colours in the shadow mask CRT.

1.7.1.5 Direct-view Storage Tubes

We know that, in raster scan display we do refreshing of the screen to maintain a screen image. The direct-view storage tubes give the alternative method of maintaining the screen image. A direct-view storage tube (DVST) uses the storage grid which stores the picture information as a charge distribution just behind the phosphor-coated screen.

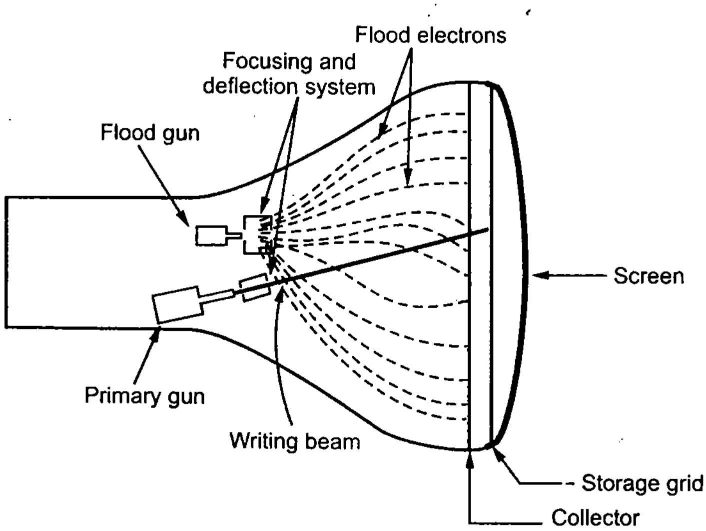


Fig. 1.29 Arrangement of the DVST

The Fig. 1.29 shows the general arrangement of the DVST. It consists of two electron guns: a primary gun and a flood gun.

A primary gun stores the picture pattern and the flood gun maintains the picture display.

A primary gun produces high speed electrons which strike on the storage grid to draw the picture pattern. As electron beam strikes on the storage grid with high speed, it knocks out electrons from the storage grid keeping the net positive charge. The knocked out electrons are attracted towards the collector. The net positive charge on the storage grid is nothing but the picture pattern. The continuous low speed electrons from flood gun pass through the control grid and are attracted to the positive charged areas of the storage grid. The low speed electrons then penetrate the storage grid and strike the phosphor coating without affecting the positive charge pattern on the storage grid. During this process the collector just behind the storage grid smooths out the flow of flood electrons.

Advantages of DVST

1. Refreshing of CRT is not required.
2. Because no refreshing is required, very complex pictures can be displayed at very high resolution without flicker.
3. It has flat screen.

Disadvantages of DVST

1. They do not display colours and are available with single level of line intensity.
2. Erasing requires removal of charge on the storage grid. Thus erasing and redrawing process takes several seconds.
3. Selective or part erasing of screen is not possible.
4. Erasing of screen produces unpleasant flash over the entire screen surface which prevents its use of dynamic graphics applications.
5. It has poor contrast as a result of the comparatively low accelerating potential applied to the flood electrons.
6. The performance of DVST is some what inferior to the refresh CRT.

1.7.1.6 Flat Panel Displays

The term flat-panel display refers to a class of video devices that have reduced volume, weight, and power requirements compared to a CRT. The important feature of flat-panel display is that they are thinner than CRTs. There are two types of flat panel displays : emissive displays and nonemissive displays.

Emissive displays : They convert electrical energy into light energy. Plasma panels, thin-film electro luminescent displays, and light emitting diodes are examples of emissive displays.

Nonemissive displays : They use optical effects to convert sunlight or light from some other source into graphics patterns. Liquid crystal display is an example of nonemissive flat panel display.

1.7.1.7 Plasma Panel Display

Plasma panel display writes images on the display surface point by point, each point remains bright after it has been intensified. This makes the plasma panel functionally very similar to the DVST eventhough its construction is markedly different.

The Fig. 1.30 shows the construction of plasma panel display. It consists of two plates of glass with thin, closely spaced gold electrodes. The gold electrodes are attached to the inner faces and covered with a dielectric material. These are attached as a series of vertical conducting ribbons on one glass plate, and a set of horizontal ribbons to the other glass plate. The space between two glass plates is filled with neon-based gas and sealed. By applying voltages between the electrodes the gas within the panel is made to behave as if it were divided into tiny cells, each one independent of its neighbours. These independent cells are made to glow by placing a firing voltage of about 120 volts across it by means of the electrodes. The glow can be sustained by maintaining a high frequency alternating voltage of about 90 volts across the cell. Due to this refreshing is not required.

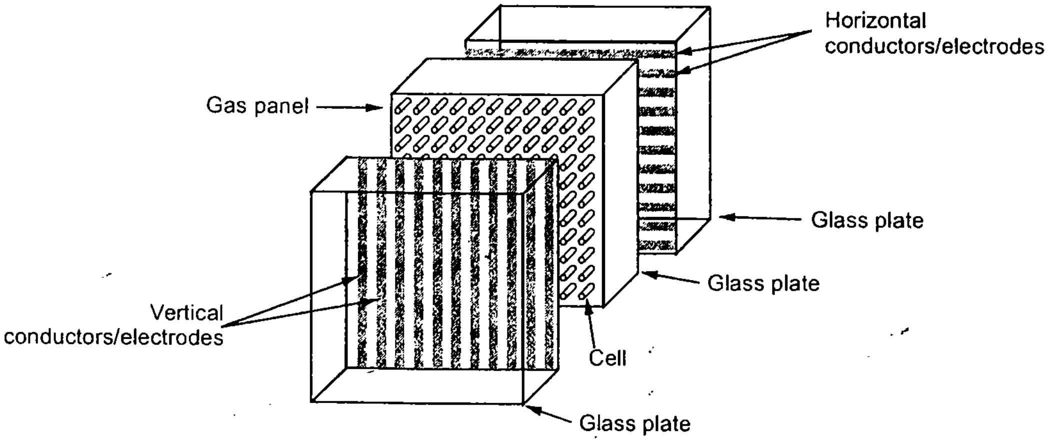


Fig. 1.30 Construction of plasma panel display

Advantages

1. Refreshing is not required.
2. Produces a very steady image, totally free of flicker.
3. Less bulky than a CRT.
4. Allows selective writing and selective erasing, at speed of about 20 μsec per cell.
5. It has the flat screen and is transparent, so the displayed image can be superimposed with pictures from slides or other media projected through the rear panel.

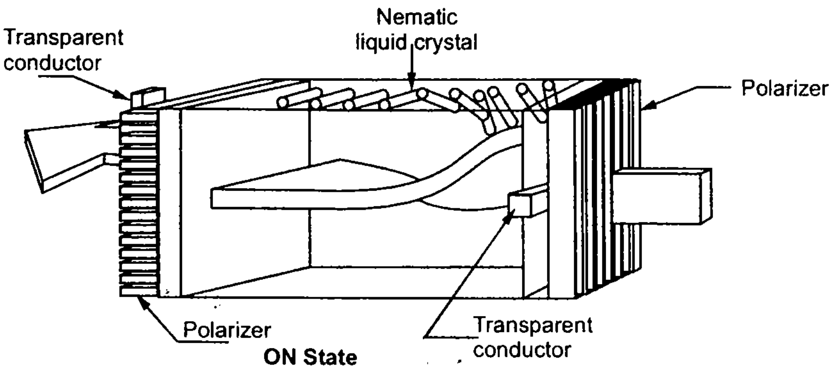
Disadvantages

1. Relatively poor resolution of about 60 dots per inch.
2. It requires complex addressing and wiring.
3. Costlier than the CRTs.

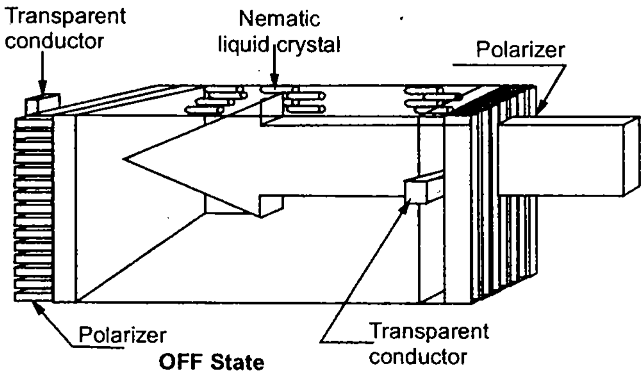
1.7.1.8 Liquid Crystal Monitors

The term liquid crystal refers to the fact that these compounds have a crystalline arrangement of molecules, yet they flow like a liquid. Flat panel displays commonly use nematic (thread like) liquid-crystal compounds that tend to keep the long axes of the rod-shaped molecules aligned.

Two glass plates, each containing a light polarizer at right angles to the other plate sandwich the liquid-crystal material. Rows of horizontal transparent conductors are built into one glass plate, and columns of vertical conductors are put into the other plate. The intersection of two conductors defines a pixel position. In the ON state, polarized light passing through material is twisted so that it will pass through the opposite polarizer. It is then reflected back to the viewer. To turn OFF the pixel, we apply a voltage to the two intersecting conductors to align the molecules so that the light is not twisted as shown in the Fig. 1.31. This type of flat panel device is referred to as a passive matrix LCD.



(a) Field effect display 'ON State'



(b) Field effect display 'OFF State'

Fig. 1.31

1.7.1.9 Important Characteristics of Video Display Devices

Persistence : The major difference between phosphors is their persistence. It decides how long they continue to emit light after the electron beam is removed. Persistence is defined as the time it takes the emitted light from the screen to decay to one-tenth of its original intensity. Lower persistence phosphors require higher refreshing rates to maintain a picture on the screen without flicker. However it is useful for displaying animations. On the other hand higher persistence phosphors are useful for displaying static and highly complex pictures.

Resolution : Resolution indicates the maximum number of points that can be displayed without overlap on the CRT. It is defined as the number of points per centimeter that can be plotted horizontally and vertically.

Resolution depends on the type of phosphor, the intensity to be displayed and the focusing and deflection systems used in the CRT.

Aspect Ratio : It is the ratio of vertical points to horizontal points to produce equal length lines in both directions on the screen. An aspect ratio of 4/5 means that a vertical line plotted with four points has the same length as a horizontal line plotted with five points.

1.7.2 Hardcopy Devices

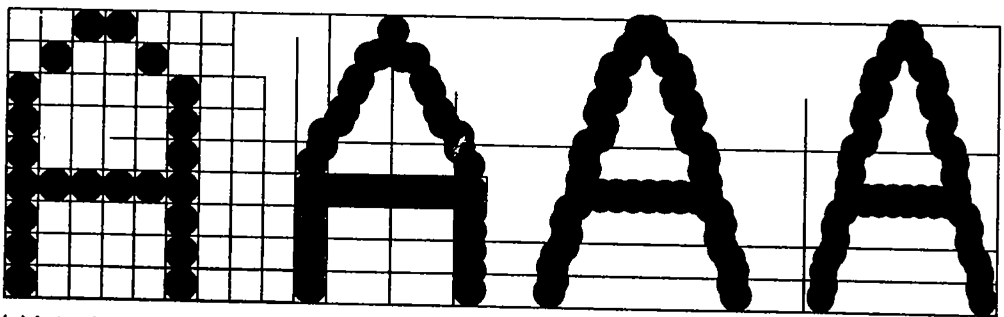
We can obtain hard-copy output for our image in several formats using printer or plotter. Therefore, printers and plotters are also called hard-copy devices. The quality of pictures obtained from a hard-copy device depends on dot size and the number of dots per inch, or lines per inch, that can be printed, i.e. it depends on the resolution of printer or plotter. Before going to see working principle of various plotters and printers we see the important characteristics of hardcopy devices.

1.7.2.1 Important Characteristics of Hardcopy Devices

Dot Size : It is the diameter of a single dot on the device's output. It is also referred to as **spot size.**

Addressability : It is the number of individual dots (not necessarily distinguishable) per inch that can be created. If the address of current dot is (x, y), then address of next dot in the horizontal direction is given as (x+1, y). Similarly, the address of next dot in vertical direction is (x, y+1).

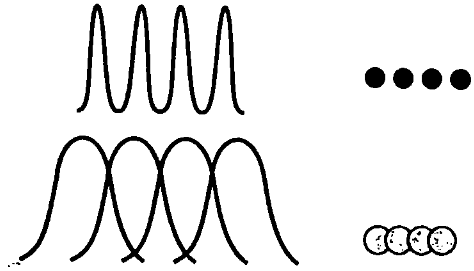
Interdot distance : It is the reciprocal of addressability. If addressability is large the interdot distance is less. The interdot distance should be less to get smooth shapes, as shown in the Fig. 1.32.



(a) Interdot spacing equal to dot size (b) Interdot spacing one half dot size (c) Interdot spacing one-thrid dot size (d) Interdot spacing one-quarter dot size

Fig. 1.32

Resolution : It is the number of distinguishable lines per inch that a device can create. It depends on a dot size and the cross-sectional intensity distribution of a spot. A spot with sharply delineated edges yields higher resolution than does one with edge that trail off, as shown in the Fig. 1.33.



(a) Spot with sharp defined edges

(b) A wider spot with less high, since the energy is spread out over a large area; its edges are not well defined

Fig. 1.33

1.7.2.2 Printers

Printers can be classified according to their printing methodology : **Impact printers** and **Non-impact printers.** Impact printers press formed character faces against an inked ribbon onto the paper. A line printer and dot matrix printer are the examples of an impact printers. Non impact printers and plotters use laser techniques, ink-jet sprays, xerographic processes, electrostatic methods, and electrothermal methods to get images onto the paper. An ink-jet printer and laser printer are the examples of non-impact printers.

Line Printers

A line printer prints a complete line at a time. The printing speed of line printer vary from 150 lines to 2500 lines per minute with 96 to 100 characters on one line. The line printers are divided into two categories : Drum printers and chain printer.

Drum Printers

A drum printers consists of a cylindrical drum. One complete set of characters is embossed on all the print positions on a line, as shown in the Fig. 1.34. The character to be printed is adjusted by rotating drum.

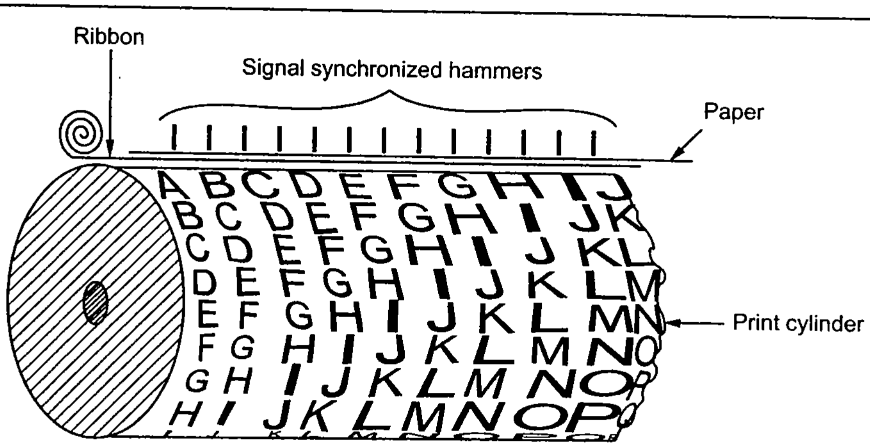


Fig. 1.34 Cylinder of a drum printer

The codes of all characters to be printed on line are transmitted from the memory of the computer to a printer memory, commonly known as **printer buffer**. This printer buffer can store 132 characters. A print drum is rotated with high speed and when printer buffer information matches with the drum character, character is printed by striking the hammer. Thus to print one line drum has to rotate one full rotation. A carbon ribbon and paper are in between the hammer and the drum therefore when hammer strikes the paper an impression is made on the backside of the paper by the ribbon mounted behind the paper. In drum printers to get good impression of the line on paper it is necessary to synchronize the movements of drum and the hammer.

Chain Printers

In these printers chain with embossed character set is used, instead of drum. Here, the character to be printed is adjusted by rotating chain. To print line, computer loads the code of all characters to be printed on line into print buffer. The chain rotated and when character specified in the print buffer appears in front of hammer, hammer strikes the carbon ribbon. A carbon ribbon is placed between the chain, paper and hammer. In this printer to get good printing quality the movement of hammer and chain must be synchronized.

Dot Matrix Printers

Dot matrix printers are also called **serial printers** as they print one character at a time, with printing head moving across a line. In dot matrix printer the print head consists of a 9×7 array of pins. As per the character definition pin are moved forward to form a character and they hit the carbon ribbon in front of the paper thereby printing that character, as shown in Fig. 1.35.

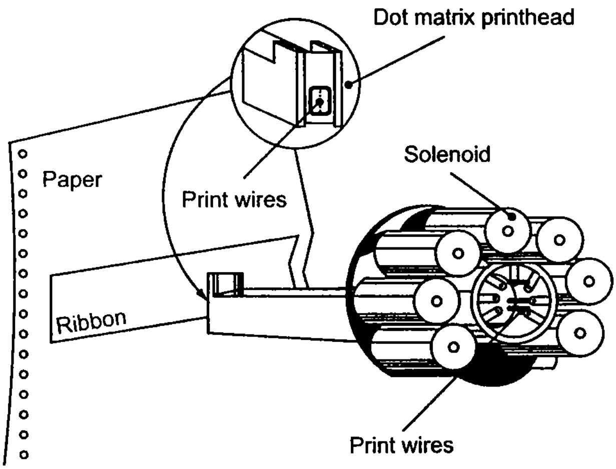
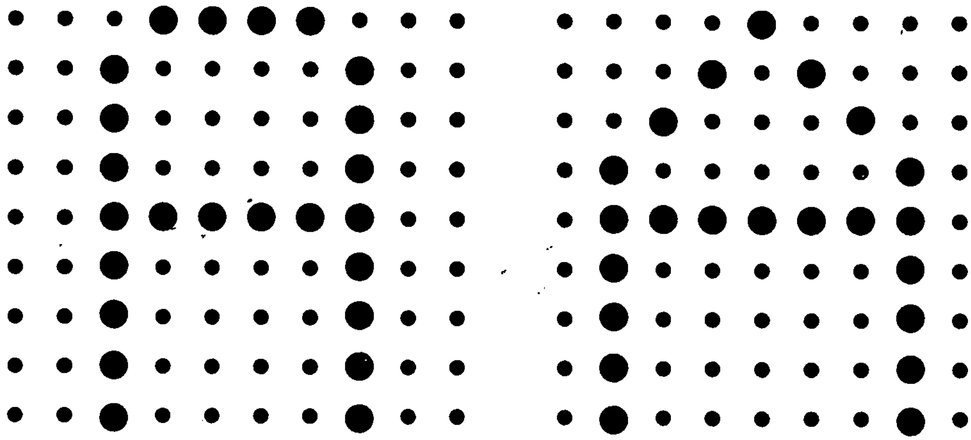


Fig. 1.35

In these printers character definition can be changed to get different font as shown in the Fig. 1.36.



(a) Dot pattern for A (b) Dot pattern for A

Fig. 1.36

An other advantage of dot matrix printers is that they can print alphabets other than English, such as Devangari, Tamil etc.

Comparison between line printer and dot matrix printer

	Line printer	Dot matrix printer
1)	Prints one line at a time.	Prints a character at a time.
2)	Characters are embossed on the drum or chain.	Characters are formed by combination of dots.
3)	Characters can not be printed with different fonts.	Characters can be printed with various fonts.
4)	Better printing quality.	Poor printing quality as characters are formed by combination of dots.
5)	Better printing speed.	Poor printing speed.
6)	Heavy duty printers.	Light duty printers.

Ink Jet Printer

An ink-jet printer places extremely small droplets of ink onto paper to create an image. If we ever look at a piece of paper that has come out of an ink-jet printer, we know that : the dots are extremely small (usually between 50 and 60 microns in diameter), so small that they are thinner than the diameter of a human hair (70 microns). The dots are positioned very precisely, with resolutions of up to 1440×720 dots per inch (dpi). The dots can have different colours combined together to create photo-quality images.

Ink jet printers print directly on paper by spraying ink through tiny nozzles as shown in the Fig. 1.37.

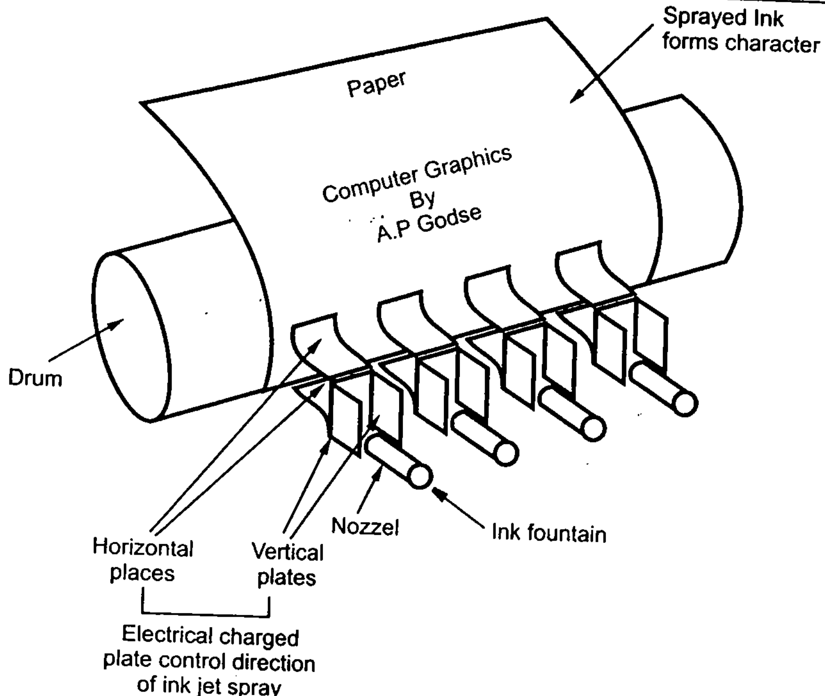


Fig. 1.37 Ink jet printer

As shown in the Fig. 1.37, the ink is deflected by an electric field with the help of horizontal and vertical charged plates to produce dot matrix patterns.

Features of ink-jet printer

1. They can print from two to four pages per minute
2. Resolution is about 360 dots per inch, therefore better printing quality is achieved.
3. The operating cost is quite low, the only part that needs replacement is the ink cartridge.
4. Colour ink jet printers have four ink nozzles with colours cyan, magenta, yellow and black, because it is possible to combine these colours to create any colour in the visible spectrum.

Laser Printer

The line, dot matrix, and ink jet printers need a head movement on a ribbon to print characters. This mechanical movement is relatively slow due to the high inertia of mechanical elements. In laser printers these mechanical movements are avoided. In these printers, an electronically controlled laser beam traces out the desired character to be printed on a photoconductive drum. The exposed areas of the drum gets charged, which attracts an oppositely charged ink from the ink toner on to the exposed areas. This image is then transferred to the paper which comes in contact with the drum with pressure and heat, as shown in the Fig. 1.38. The charge on the drum decides the darkness of the print. When charge is more, more ink is attracted and we get a dark print.

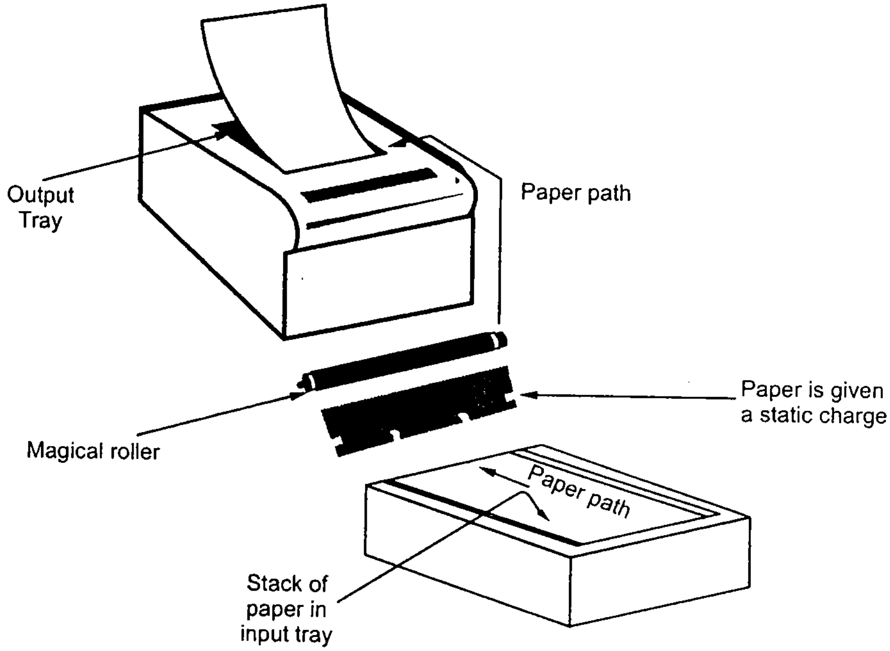


Fig. 1.38 Laser-printer

A colour laser printer works like a single colour laser printer, except that the process is repeated four times with four different ink colours : Cyan, magenta, yellow and black. Laser printers have high resolution from 600 dots per inch upto 1200 dots per inch. These printers print 4 to 16 page of text per minute. The high quality and speed of laser printers make them ideal for office environment.

Advantages of Laser printer

The main advantages of laser printers are speed, precision and economy. A laser can move very quickly, so it can "write" with much greater speed than an ink-jet. Because the laser beam has an unvarying diameter, it can draw more precisely, without spilling any excess ink. Laser printers tend to be more expensive than ink-jet printers, but it doesn't cost as much to keep them running. Its toner power is cheap and lasts for longer time.

Thermal Transfer Printer

In thermal transfer printer, wax paper and plain paper are drawn together over the strip of heating nibs. The heating nibs are selectively heated to cause the pigment transfer. In case of colour thermal transfer printers, the wax paper is placed on a roll of alternating, cyan, magenta, yellow and black strips, each of a length equal to the paper size. It is possible to create one colour hardcopy with less than 1 minute. This is possible because the material used to manufacture nib heats and cools very rapidly. Morden thermal transfer printers